the light string modes in this situation. The result obtained is the 1061 times the Planck length quoted earlier. If the three familiar spatial dimensions are in fact circular and string theory is right, astronomers using vastly different (and currently nonexistent) equipment, in principle, should be able to measure the extent of the heavens with heavy wound-string modes and find a result that is the reciprocal of this huge distance. It is in this sense that we can think of the universe as being either huge, as we normally do, or terribly minute. According to the light string modes, the universe is large and expanding; according to the heavy modes it is tiny and contracting. There is no contradiction here; instead, we have two distinct but equally sensible definitions of distance. We are far more familiar with the first definition due to technological limitations, but, nevertheless, each is an equally valid concept.

Now we can answer our earlier question about big humans in a little universe. When we measure the height of a human and find six feet, for instance, we necessarily use the light string modes. To compare their size to that of the universe, we must use the same measuring procedure and, as above, this yields 15 billion light-years for the size of the universe, a result that is much larger than six feet. Asking how such a person can fit into the “tiny” universe as measured by the heavy string modes is asking a meaningless question—it’s comparing apples and oranges. Since we now have two concepts of distance—using light or heavy string probes—we must compare measurements made in the same manner.

A Minimum Size

It’s been a bit of a trek, but we are now set for the key point. If one does stick to measuring distances “the easy way”—that is, using the lightest of the string modes instead of the heavy ones—the results obtained will always be larger than the Planck length. To see this, let’s think through the hypothetical big crunch for the three extended dimensions, assuming them to be circular. For argument’s sake, let’s say that at the beginning of our thought experiment, unwound string modes are the light ones and by using them it is determined that the universe has an enormously large radius and that it is shrinking in time. As it shrinks, these unwound modes get heavier and the winding modes get lighter. When the radius shrinks all the way to the Planck